# How Much of AI-Text Detection Accuracy Is Real? A Dataset-Dependent Reliability Audit of DAIGT V2 and HC3

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***Abstract*—Detectors of AI-generated text routinely report accuracy at or above 99% on benchmarks such as DAIGT V2 and HC3, but this in-distribution number says little about real-world reliability. We decompose reported accuracy into three components, cross-dataset generalization, dependence on surface artifacts, and robustness to cheap adversarial attack, using a five-model pipeline (three classical baselines and fine-tuned BERT and DeBERTa) at full corpus scale. A contamination audit finds HC3 carries substantial internal duplication (7.16%) essentially absent from DAIGT V2 (0.01%). Strict one-way cross-dataset transfer, evaluated across three seeds, shows substantial degradation from both datasets in both directions. An artifact-cleaning ablation, run both zero-shot and via full retraining, finds a real negative accuracy delta for HC3 but not DAIGT V2, consistent with HC3's stronger surface artifact, a whitespace-before-punctuation cue present in 88.7% of human answers versus 0.28% of ChatGPT answers. An adversarial evaluation across typo injection, homoglyph substitution, and back-translation then finds HC3 accuracy collapses to near-chance under a typo attack, as low as 0.36 F1 from an in-domain 0.998, while DAIGT V2 stays comparatively robust, three independent experiments converging on the same conclusion for HC3. These results support a dataset-dependent verdict, HC3's headline accuracy substantially overstates real-world reliability while DAIGT V2's does so only modestly, a distinction invisible from in-distribution accuracy alone.**

***Index Terms*—AI-generated text detection, benchmark reliability, adversarial robustness, artifact leakage, cross-dataset generalization**

## I. Introduction

The proliferation of capable large language models has made distinguishing human-written from machine-generated text a practical concern across education, journalism, and content moderation, and a substantial body of work has responded by fine-tuning transformer classifiers on paired human/AI corpora. Two of the most widely used benchmarks for this task, DAIGT V2 [1] and HC3 [2], have each been the subject of dozens of published detection studies, and the reported results are, almost without exception, striking, in-distribution F1 at or above 97%, with several studies exceeding 99% [2]. Read at face value, these numbers suggest AI-generated text detection is close to solved on the domains these datasets represent.

This paper starts from the observation that the apparent solvedness is an artifact of how these benchmarks are typically evaluated, not evidence that detectors have learned to recognize AI-generated language in a way that generalizes. Prior work each undermines part of the 99% headline number from a different angle, cross-dataset evaluations show sharp collapses across distribution shift [3], documented surface artifacts allow detectors to score well without learning language patterns [4], [5], and adversarial perturbation studies show near-99%-in-domain detectors collapsing under simple attacks [6]. No prior study, to our knowledge, combines all three lines of evidence, strict one-way cross-dataset transfer, an artifact-cleaning ablation, and adversarial robustness, on exactly these two benchmarks within one directly comparable design.

This project builds on a validated five-model detection pipeline (three classical baselines, fine-tuned BERT and DeBERTa) evaluated at full corpus scale (34,994 DAIGT V2 rows, 53,806 HC3 rows), with a reproducible split protocol and a seed-robustness study already in place. Extending this infrastructure surfaces an additional finding in its own right, BERT's seed-to-seed fine-tuning instability shrinks by roughly 24-fold moving from a 6,000-row sample to the full corpus (0.0267 to 0.0011 F1 spread), a scale-dependence not stated by the foundational fine-tuning-instability literature [7], [8].

Our contribution is threefold. First, a full-corpus contamination audit of both datasets finds HC3 carries substantial internal duplication (7.16% of 85,449 rows) essentially absent from DAIGT V2 (0.01%), a direct dataset-versus-dataset comparison absent from prior single-dataset audits. Second, we establish that BERT's fine-tuning instability is itself scale-dependent. Third, combining cross-dataset transfer, an artifact-cleaning ablation, and an adversarial robustness evaluation, we quantify how much of the widely reported 99% detection accuracy survives contact with a different distribution, a cleaned evaluation set, and a cheap adversarial attack, addressing directly the question raised but not fully answered by the closest prior work [3]. The results are dataset-dependent, HC3's headline accuracy substantially overstates real-world reliability, while DAIGT V2's does so only modestly.

## II. Related Work

Across the broader literature on both datasets, essentially every study evaluating purely in-distribution reports F1 in the 97-100% range [2], [9], [10], [11], [12], a saturation meaning another in-distribution accuracy number does not by itself constitute a novel contribution. A separate thread shows this apparent solvedness is fragile. Antoun et al. [6] attack HC3-trained detectors with misspellings and homoglyphs, a detector scoring 99.88 F1 in-domain falls to 33.57% raw accuracy on a hand-written adversarial set. Su et al. [9] show reframing detection from question-answering to semantic-invariant tasks drops accuracy from 99.82% to roughly 91.7%. Borile and Abrate [13] find that ablating around twenty feed-forward neurons, about 0.05% of a BERT-based detector, improves out-of-distribution accuracy by up to 6.9 points, mechanistic evidence that a measurable share of in-distribution performance is carried by dataset-specific, non-generalizing features. A third thread documents concrete surface artifacts, Tian et al. [4] discover an extra space before punctuation in HC3-English human answers absent from ChatGPT answers, Baidya et al. [5] independently document a length confound in the same dataset, and Ardeshirifar [14] standardizes punctuation and contractions for the same underlying reason, three independent groups converging on the conclusion that reported HC3 accuracy is partly a function of collection artifacts rather than language understanding. Park et al. [15] extend this further, showing HC3-trained detectors latch onto prompt-specific collection shortcuts and constructing adversarial instructions that exploit them directly. On the classical-baseline side, Kubrusly et al. [16] report strong bag-of-words and TF-IDF results on HC3 without any transformer, the natural comparison point for the classical models in our own pipeline.

The work closest to ours is Alikhanov et al. [3], who merge HC3 and DAIGT V2 into one 124,195-sample corpus and evaluate under a topic-held-out split, reporting substantially lower accuracy (82.87-88.86%) than in-distribution numbers reported elsewhere. Their design asks whether one model can handle both datasets at once, whereas our strict one-way cross-dataset transfer asks whether detection knowledge acquired on one corpus carries to a genuinely disjoint one, and our artifact-cleaning and adversarial-robustness results decompose *why* transfer fails rather than only measuring that it does.

## III. Methods

### A. Datasets

DAIGT V2 [1] contains 44,868 argumentative student essays, 27,371 human-written (PERSUADE 2.0) and 17,497 machine-generated across a mixture of GPT-3.5, Llama-2, Mistral-7B, Falcon-180B, and other 2023-era models, a 61%/39% human/AI split. HC3 [2] contains question-answer pairs across five English sources, contrasting human answers against GPT-3.5-Turbo-0301 responses, 85,449 rows total. Both were balanced for full-scale training (34,994 DAIGT V2 rows, 53,806 HC3 rows).

A duplicate-detection audit finds DAIGT V2 has essentially no internal duplication, six rows across six groups (0.01%), zero cross-label duplicates. HC3 shows a substantially different picture, 6,118 duplicate rows across 5,986 groups (7.16%), plus 476 rows nominally labeled ChatGPT-authored that are collection artifacts, text such as API failure messages stored under the answer field. These figures corroborate a smaller, directly inspectable leakage check at 1,200-row test scale, eight leaked rows (0.67%), rescoring on the leakage-excluded subset changes F1 by at most 0.0010 with no ranking change, so the leakage's effect on the reported metric is smaller than ordinary seed-to-seed noise even though the leakage itself is real.

### B. Models

Five classifiers span the classical-to-transformer spectrum, Naive Bayes, Logistic Regression, and a linear SVM over bag-of-words or TF-IDF features, and fine-tuned `bert-base-uncased` and `microsoft/deberta-v3-base`. Transformers were swept over learning rate $\{2e{-}5, 3e{-}5\}$, batch size $\{16, 32\}$, and weight decay $\{0.01, 0.1\}$, best configuration selected by validation F1, with 128-token truncation inherited from the originating course specification, an open limitation discussed in Section V. A subset of winning configurations was re-run at seeds 123 and 456, in addition to the default 42, to establish seed-to-seed variance.

### C. Cross-dataset transfer

Each transformer is trained on one dataset's training split and evaluated on the other dataset's held-out test split, in both directions, across all three seeds, alongside the matched in-domain baseline. This is a strict one-way transfer design, distinct from Alikhanov et al.'s [3] merge-and-topic-split approach, transferring asks whether knowledge of one dataset generalizes to the other, a harder question than whether one model can handle both.

### D. Artifact-cleaning ablation

Two variants were run per model and dataset, a zero-shot pass re-scoring the existing raw-trained checkpoint on cleaned test text only, and a full ablation retraining from scratch on cleaned train/validation/test data at the same fixed configuration. Cleaning removes the whitespace-before-punctuation artifact for HC3 and the emoji/pictograph signal for DAIGT V2 (present in 3.2% of AI-labeled versus 0% of human-labeled DAIGT V2 essays), plus per-row length-matching for the full-retrain variant.

### E. Adversarial robustness

Test text only, never training data, is perturbed via character-level typo injection and homoglyph substitution (Latin-to-Cyrillic visual lookalikes) at 1%, 5%, and 10% of eligible characters, and a single back-translation (English-German-English) paraphrase pass, then scored against the same in-domain checkpoints. All experiments use bootstrap-appropriate reporting, no parametric significance tests at $n \leq 30$, given the small seed count.

## IV. Results

### A. In-distribution baseline

All five classifiers saturate above 0.94 F1 on both datasets, both transformers exceed 0.99 (DeBERTa reaches 0.9949 on DAIGT V2 and 0.9980 on HC3, the highest number in this project). This comparison is a methodological sanity check, not the paper's central claim, in-distribution saturation on these two benchmarks is already well established.

TABLE I
STRICT ONE-WAY TRANSFER, MEAN OVER 3 SEEDS

| Trained on | Model | In-dom. | Cross-dom. | Gap |
|---|---|---|---|---|
| DAIGT V2 | BERT | 0.9921 | 0.7902 | 0.2019 |
| DAIGT V2 | DeBERTa | 0.9930 | 0.9096 | 0.0833 |
| HC3 | BERT | 0.9930 | 0.8311 | 0.1619 |
| HC3 | DeBERTa | 0.9970 | 0.8512 | 0.1458 |

TABLE II
F1 UNDER 10% TYPO INJECTION, WORST CASE PER MODEL

| Dataset | Model | In-dom. | Attacked | Drop |
|---|---|---|---|---|
| DAIGT V2 | BERT | 0.9916 | 0.7966 | 0.195 |
| DAIGT V2 | DeBERTa | 0.9949 | 0.9590 | 0.036 |
| HC3 | BERT | 0.9945 | 0.3746 | 0.620 |
| HC3 | DeBERTa | 0.9980 | 0.3644 | 0.634 |

### B. *Scale-dependent seed instability*

Re-running the winning DAIGT V2 BERT configuration at three seeds yields test F1 values of 0.9916, 0.9927, and 0.9920, a spread of 0.0011, versus 0.0267 at the smaller 6,000-row scale, an approximately 24-fold reduction. Established instability results [7], [8] document that fine-tuning outcomes vary across seeds but do not state that this instability is itself scale-dependent, a genuine refinement rather than a replication.

### C. *Cross-dataset transfer*

Every cell in Table I collapses substantially relative to its in-domain baseline. The direction is mixed by model, DAIGT-V2-trained BERT transfers worse to HC3 than the reverse, while HC3-trained DeBERTa transfers worse to DAIGT V2 than the reverse, so transfer alone does not uniformly single out HC3.

### D. *Artifact-cleaning ablation*

Under full retraining on cleaned data, the effect differs sharply by dataset. On DAIGT V2, cleaning the emoji signal produces no meaningful change, BERT moves from 0.9916 to 0.9936 and DeBERTa from 0.9917 to 0.9943, both within the seed-noise band above. On HC3, where the whitespace-before-punctuation artifact is far stronger, present in 88.7% of human rows versus 0.28% of ChatGPT rows, cleaning produces a real negative delta for both models, BERT from 0.9917 to 0.9916 and DeBERTa from 0.9973 to 0.9962. The zero-shot pass (no retraining) shows a similar but smaller HC3 DeBERTa effect (0.9973 to 0.9906) and no DAIGT V2 effect, consistent with the DAIGT V2 artifact touching only 3.2% of AI-labeled rows overall.

### E. *Adversarial robustness*

Table II shows the sharpest split in the paper. DAIGT V2 stays comparatively robust across every attack and strength tested, HC3 collapses to near-chance under moderate typo injection. Homoglyph substitution shows the same pattern, less extreme, back-translation is markedly milder than either character-level attack across every cell (largest drop 0.021), consistent with it being a semantically faithful paraphrase rather than a character-level corruption that directly targets the artifacts documented above.

## V. DISCUSSION AND CONCLUSION

Taken individually, each experiment qualifies rather than overturns the field's dominant narrative. Taken together, they support a more specific claim, the $\sim$99% headline accuracy reported throughout the literature on these benchmarks is real in the narrow sense that it reproduces reliably in-distribution, but it decomposes into components of sharply different reliability by dataset. The artifact-cleaning and adversarial results point unambiguously the same direction, HC3 is the only dataset where the cleaning retrain produces a real negative delta for both models, and HC3 is the dataset where adversarial attacks collapse accuracy to near-chance. That two methodologically distinct experiments, one retrains on a surface-cue-removed corpus, the other perturbs test text at inference time only, converge on the same dataset is evidence the underlying phenomenon, reliance on non-semantic, HC3-specific surface signal, is real rather than an artifact of either design choice alone. DAIGT V2's comparative robustness across both experiments does not mean its classification reflects deeper language understanding, cross-dataset transfer shows DAIGT-V2-trained models collapse on HC3 just as HC3-trained models collapse on DAIGT V2, so neither model has learned a domain-general notion of AI-generated text, only that whatever DAIGT V2 classification depends on is not the same brittle, character-level cue that HC3 classification depends on.

We state one limitation explicitly rather than leave it implicit. The 128-token input truncation inherited from this project's originating course specification discards a median of 69% of each DAIGT V2 essay's content, and does so asymmetrically, human essays run measurably longer than AI essays in our sample (median 448.5 versus 398.5 tokens), so truncation removes proportionally more human content than AI content. Whether DAIGT V2's comparative robustness survives a longer sequence length is an open question this paper does not resolve, and is, in our view, the single most important remaining one.

This paper does not build a better detector, nor attempt to. Its contribution is diagnostic, a reproducible method for asking, of any reported AI-text-detection accuracy number, how much of it survives a different data distribution, a cleaned evaluation set, and a cheap adversarial attack. Applied to two of the field's most widely used benchmarks, the answer is dataset-dependent and, for HC3, large enough that the single headline number materially misrepresents real-world reliability.

## DATA AND CODE AVAILABILITY

Both datasets (DAIGT V2, HC3) are public; code is released for reproduction. The link is withheld for blind review.

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